



Eventard

Campus event board with live RSVPs and Add to Google Calendar

A 300-level project for the University of Uyo. Students browse events, RSVP and save them to their calendar. Student Affairs publishes events that appear on every open board within seconds.

38

seeded events

50

users (2 admins)

1092

RSVPs

Contents

1. Overview
2. How the core logic works
3. Architecture and data model
4. Screen walkthrough
5. Mobile view
6. Running locally
7. Testing
8. Deployment
9. Five minute presentation script

Demo logins: student@uniuyo.edu.ng / student123 and admin@uniuyo.edu.ng / admin123. Screens captured with the clock frozen at 2026-10-05.

1. Overview

Students miss campus events because announcements are scattered across WhatsApp groups and notice boards. Eventard puts them on one board.

Who	What they can do
Student	Browse upcoming and past events, search and filter by category, RSVP or cancel, see the live count and who is going, add an event to Google Calendar, see their own RSVPs.
Admin (Student Affairs)	See dashboard numbers, publish events with a banner (upload or library), edit, cancel, restore or delete events, and see the full attendee list.

Features

- Live RSVP count without a page reload (server re-render every 4 to 6 seconds).
- Capacity limits that cannot be overbooked, even when students click at the same moment.
- Add to Google Calendar with no API key.
- Full, almost full, live, ended and cancelled states.
- Inline form validation with Zod.
- Works on phones.

Design

The interface follows the visual style of Claude: warm ivory background, serif headings (Source Serif 4), a clean sans body (Instrument Sans), soft borders and a terracotta accent. It is a normal web app, not a chatbot.

2. How the core logic works

All business rules are pure functions in `src/lib/`. They take "now" as an argument, so the tests and the demo can freeze the clock with `EVENTARD_TODAY=YYYY-MM-DD` (the clock is then 10:00 Lagos time on that day).

Event phase

```
eventPhase(event, now)
  cancelled  if status is cancelled
  upcoming   if now < starts_at
  live       if starts_at <= now < ends_at
  ended      otherwise
```

Can this student RSVP?

```
rsvpState(event, goingCount, hasRsvp, now)
  cancelled or ended  -> closed
  hasRsvp             -> "going" (button cancels)
  live                -> closed for new RSVPs
  capacity reached    -> "full"
  otherwise           -> "open"
```

No overbooking

`toggleRsvp` runs in a database transaction. It first runs `SELECT ... FOR UPDATE` on the event row, so two requests for the same event wait for each other. It then counts RSVPs, applies `rsvpState` and inserts or deletes. A unique constraint on `(event_id, user_id)` stops duplicates. The integration test fires 10 RSVPs at a 3 seat event at once and checks exactly 3 succeed.

Add to Google Calendar

```
https://calendar.google.com/calendar/render?action=TEMPLATE
&text=Code%20and%20Coffee%3A%20Build%20a%20Web%20App%20with%20Next.js
&dates=20261006T130000Z/20261006T160000Z
&details=...&location=ICT%20Centre%2C%20Lab%202%2C%20Town%20Campus
```

Times are stored as `timestampz`. `toGoogleDate` converts to UTC and strips punctuation, so 2:00 pm Lagos becomes 130000Z. Every text field goes through `encodeURIComponent`.

Live updates

A tiny client component, `LiveRefresh`, calls `router.refresh()` on a timer while the tab is visible. Next.js re-renders the Server Components and sends only the changed parts, so the count and new events appear without a reload. Server Actions also call `revalidatePath` so the person who clicked sees the change at once.

Sessions

Passwords are hashed with `bcryptjs`. On sign in the server sets a JWT signed with HS256 (jose) in an HTTP-only, `SameSite=Lax` cookie. `src/proxy.ts` redirects visitors without a valid token, and every page and Server

Action checks the session again on the server.

3. Architecture and data model

Browser

Server-rendered pages, small client components for forms, RSVP and live refresh.

Next.js16 (App Router)

Server Components read data. Server Actions change it. proxy.ts guards routes. /api/health pings the database.

PostgreSQL

Drizzle ORM with versioned SQL migrations from drizzle-kit.

Folder	Contents
src/app	Routes: /login, / (board), /events/[id], /my, /admin, /admin/events/new, /admin/events/[id]/edit, /api/health, /api/images/[id]
src/app/actions	Server Actions: login, logout, RSVP, create, update, cancel, delete
src/lib	Pure rules (events.ts, calendar.ts, today.ts, format.ts), validation, auth, queries
src/db	Drizzle schema, connection pool, seed data
drizzle/	Generated SQL migrations
tests/, e2e/	Vitest unit and integration tests, Playwright end-to-end tests

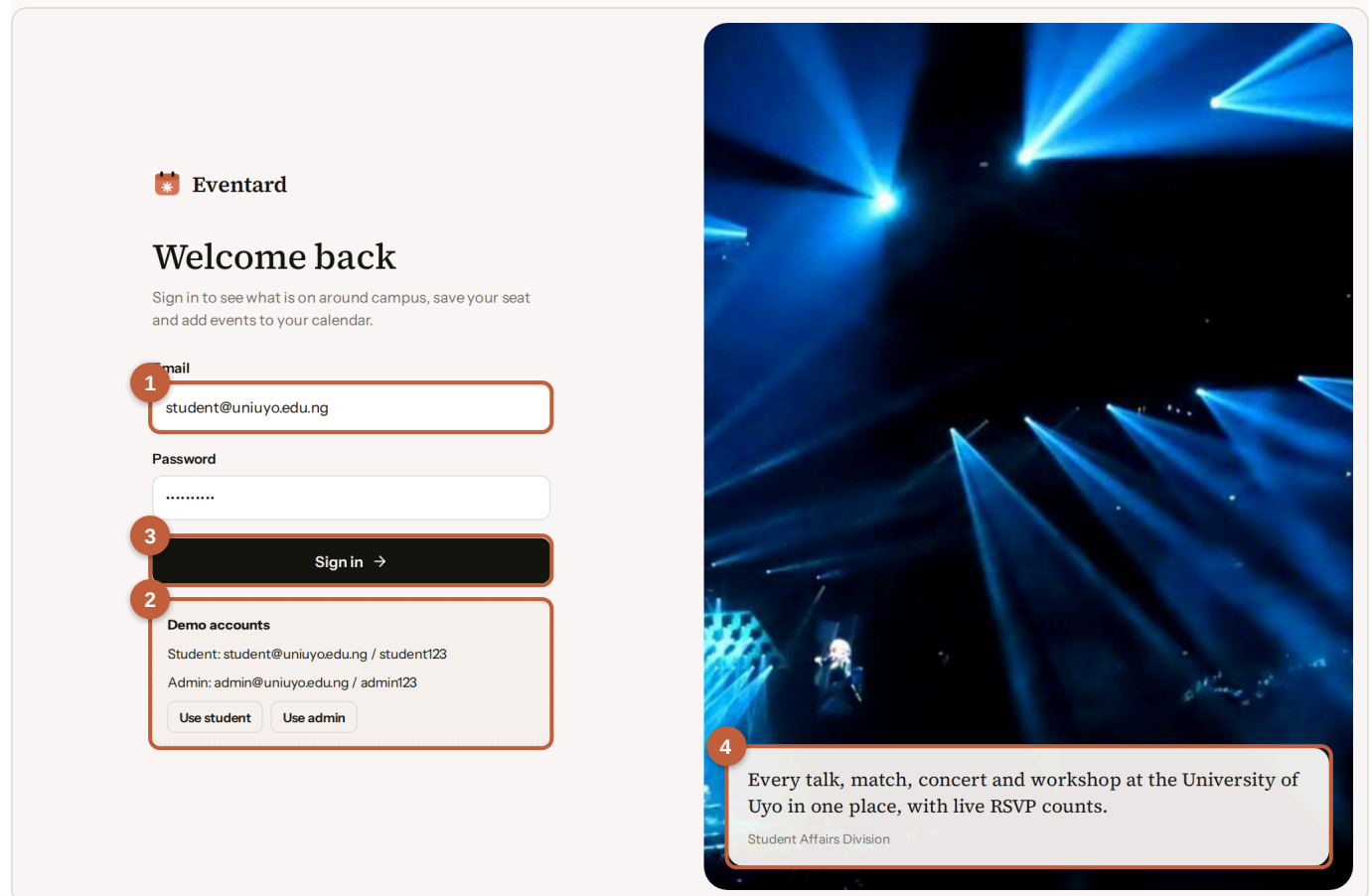
Tables

Table	Columns	Rules
users	id, name, email, password_hash, role (student or admin), department, level, created_at	email unique
events	id, title, description, category, venue, starts_at, ends_at, banner_url, capacity, status, created_by, created_at	ends_at > starts_at, capacity null or positive, created_by references users
rsvps	id, event_id, user_id, created_at	unique (event_id, user_id), cascades on delete
uploads	id (uuid), mime, data (bytea), uploaded_by, created_at	Banners uploaded by admins, served by /api/images/[id]

Uploaded banners are kept in Postgres rather than on disk because Railway containers have no persistent file system.

4. Screen walkthrough

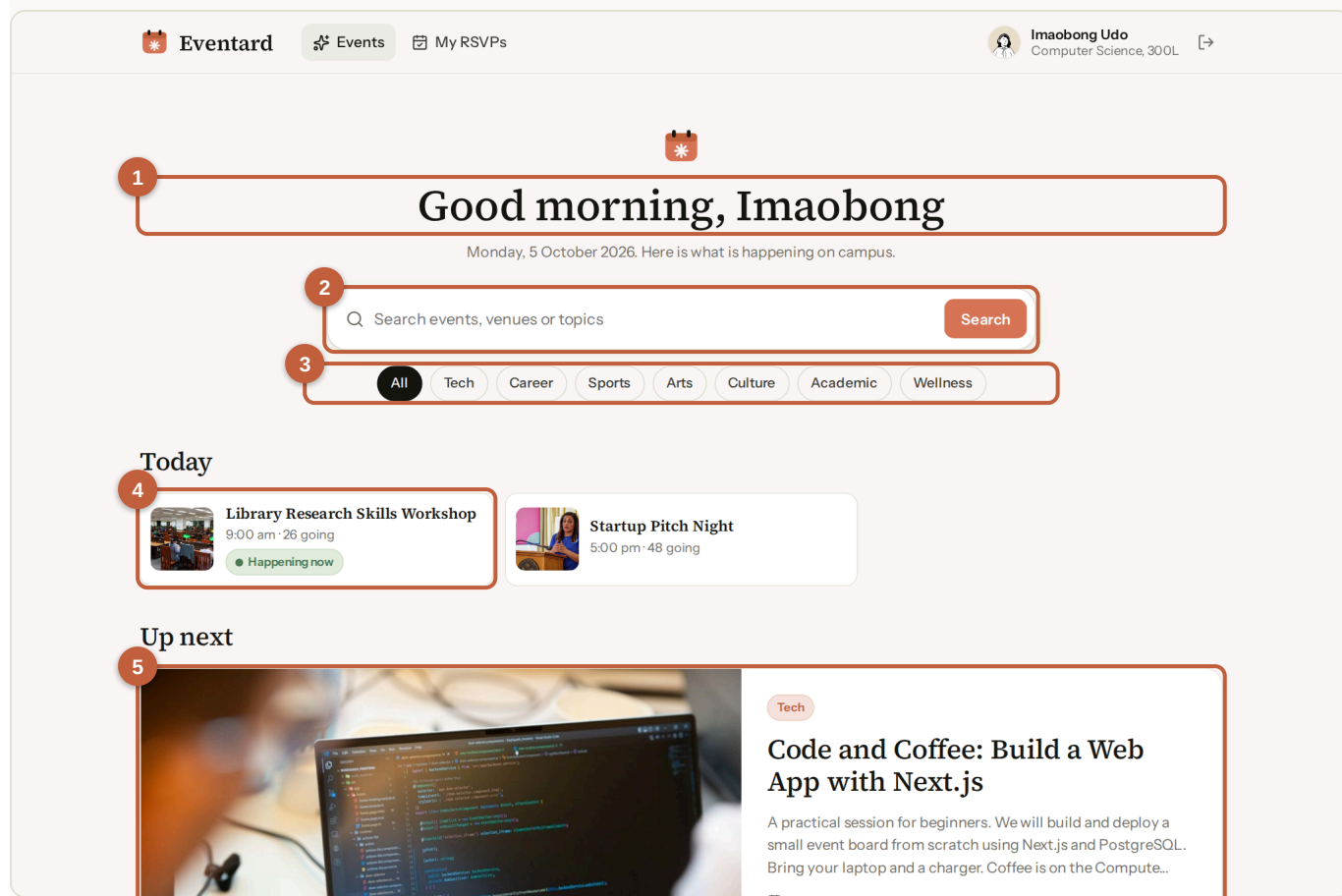
Sign in



Sign-in page with the demo login prefilled.

- 1 Email field, prefilled with the demo student account so the presenter can sign in with one click.
- 2 Demo accounts panel. The Use admin button swaps the fields to the Student Affairs admin login.
- 3 Sign in posts to a Server Action that checks the bcrypt hash and sets a signed JWT in an HTTP-only cookie.
- 4 Brand panel with a real campus concert photo (Creative Commons, stored locally).

Event board



The board as the demo student sees it.

- 1 Personal greeting using the student's first name and the campus time of day (Africa/Lagos).
- 2 Search box. Matches event titles, venues and descriptions with a case-insensitive query.
- 3 Category filter chips. The active chip is dark; the filter is kept in the URL so it can be shared.
- 4 Today strip: events on the current campus day with their start time and live RSVP count.
- 5 Up next: the nearest upcoming event, shown large so the demo starts from it.

Upcoming events grid

Coming up

1

Upcoming

Past

2

OCT 7



Career ✓ Going

Career Fair 2026: Meet 30 Employers

🕒 Wed, 7 Oct, 9:00 am

📍 Main Auditorium, Main Campus

4



46 going

3

OCT 8



Tech ✓ Going

Tiny workshop

🕒 Thu, 8 Oct, 11:00 am

📍 Lab 1



3 going

of 3

OCT 8



Sports

Inter-Faculty Basketball Semi Final

🕒 Thu, 8 Oct, 4:00 pm

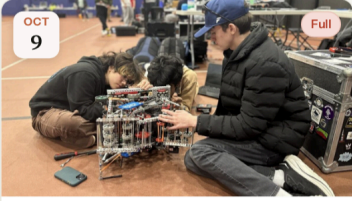
📍 Indoor Sports Hall



33 going

of 200

OCT 9



Tech

Robotics Club Open Day

🕒 Fri, 9 Oct, 11:00 am

📍 Faculty of Engineering Workshop



45 going

of 45

OCT 10



Wellness

Mental Health Awareness Talk

🕒 Sat, 10 Oct, 1:00 pm

📍 Faculty of Social Sciences Hall



0 going

OCT 11



Arts ✓ Going

Afrobeats Live: Campus Concert

🕒 Sun, 11 Oct, 6:00 pm

📍 Town Campus Field



47 going

Event cards with status flags and attendee avatars.

- 1 Upcoming and Past tabs. Past events keep their RSVP history but cannot be booked.
- 2 Date badge in campus time.
- 3 Status flag computed by eventPhase and isAlmostFull: Full, a few spots left, Happening now or Cancelled.
- 4 Avatar stack of the latest attendees with the total going. Avatars are generated locally with DiceBear.

Event page

Eventard

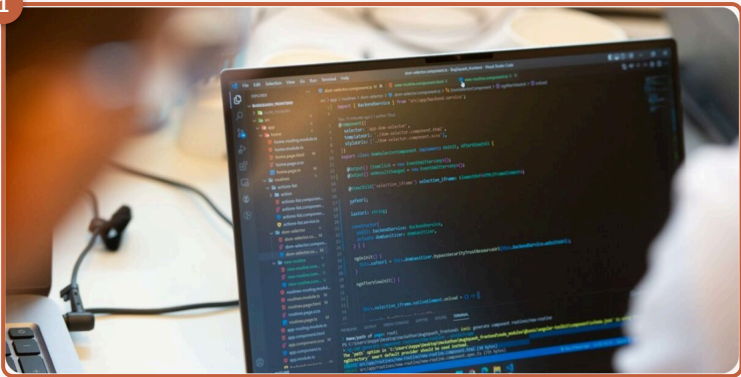
Events

My RSVPs

Imaobong Udo
Computer Science, 300L

All events

1



Tech Tomorrow

Code and Coffee: Build a Web App with Next.js

2

Tuesday, 6 October 2026

Date

2:00 pm to 5:00 pm

West Africa Time

ICT Centre, Lab 2, Town Campus

Venue

60 seats

Capacity

Live count

60 seats

37 students going

4 + RSVP to this event

5 Add to Google Calendar

Who is going

Halima Abubakar

Ruth Ekong

Daniel Ebong

Ngozi Obi

and 33 others

A practical session for beginners. We will build and deploy a small event board from scratch

Event page before the student RSVPs.

- 1 Event banner.
- 2 Date, time range (West Africa Time), venue and capacity.
- 3 Live count. The page calls `router.refresh()` every 4 seconds so other students' RSVPs show up without a reload.
- 4 RSVP button. It runs a Server Action that locks the event row, re-checks capacity and inserts the RSVP.
- 5 Add to Google Calendar. A plain link with the title, UTC dates, description and venue URL-encoded. No Google API key is needed.

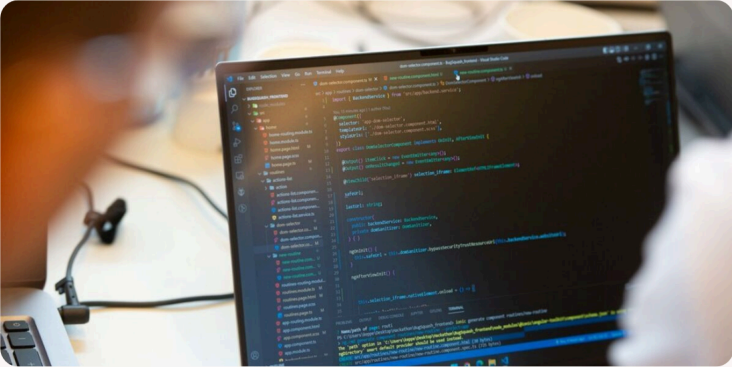
After RSVP

 **Eventard**

EventsMy RSVPs

Imaobong Udo
Computer Science, 300L

← All events



TechTomorrow

Code and Coffee: Build a Web App with Next.js

 **Tuesday, 6 October 2026**
Date

 **2:00 pm to 5:00 pm**
West Africa Time

 **ICT Centre, Lab 2, Town Campus**
Venue

 **60 seats**
Capacity

A practical session for beginners. We will build and deploy a small event board from scratch

Live count60 seats

138 students going

3

2

✓ You are going. Tap to cancel!

Add to Google Calendar

Who is going

 Halima Abubakar

 Ruth Ekong

 Daniel Ebong

 Imaobong Udo

and 34 others

The same page right after clicking RSVP.

- 1 The count went from 37 to 38 straight after the click.
- 2 The button turns green. Tapping it again cancels the RSVP and frees the seat.
- 3 Capacity bar: 38 of 60 seats taken.

My RSVPs

 **Eventard**

Events

My RSVPs

 **Imaobong Udo**
Computer Science, 300L

1

My RSVPs

6 upcoming, 4 attended. Only you can see this list.

Find more events

Upcoming

OCT 5



Career

3

Going

Startup Pitch Night

Mon, 5 Oct, 5:00 pm

Entrepreneurship Centre Hall

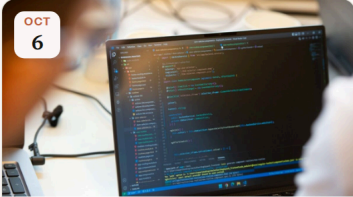
48 going

of 120

2

Add to Google Calendar

OCT 6



Tech

✓

Going

Code and Coffee: Build a Web App with Next.js

Tue, 6 Oct, 2:00 pm

ICT Centre, Lab 2, Town Campus

38 going

of 60

Add to Google Calendar

OCT 7



Career

✓

Going

Career Fair 2026: Meet 30 Employers

Wed, 7 Oct, 9:00 am

Main Auditorium, Main Campus

46 going

Add to Google Calendar

OCT 8



Full

OCT 11



OCT 16



The student's own bookings.

- 1 My RSVPs only ever queries rows where user_id is the signed-in student.
- 2 A calendar button under each upcoming booking.
- 3 Going badge.

Admin dashboard

 Eventard

Events

Dashboard

 Dr. Uwem Ekpenyong
Student Affairs

Events dashboard

Publish events and watch RSVPs come in.

24
Upcoming events

10
In the next 7 days

587
RSVPs for upcoming events

2
Upcoming events at capacity

+ New event

Upcoming events

Upcoming

Past

EVENT	WHEN	RSVPS	STATUS	
 Library Research Skills Workshop University Library Seminar Room	Mon, 5 Oct 9:00 am	26 / 30 	Live	View Edit
 Startup Pitch Night Entrepreneurship Centre Hall	Mon, 5 Oct 5:00 pm	48 / 120 	Open	View Edit
 Code and Coffee: Build a Web App with Next.js ICT Centre, Lab 2, Town Campus	Tue, 6 Oct 2:00 pm	38 / 60 	Open	View Edit
 Career Fair 2026: Meet 30 Employers Main Auditorium, Main Campus	Wed, 7 Oct 9:00 am	46 	Open	View Edit
 Tiny workshop Lab 1	Thu, 8 Oct 11:00 am	3 / 3 	Full	View Edit
 Inter-Faculty Basketball Semi Final Indoor Sports Hall	Thu, 8 Oct 4:00 pm	33 / 200 	Open	View Edit
 Robotics Club Open Day Faculty of Engineering Workshop	Fri, 9 Oct 11:00 am	45 / 45 	Full	View Edit

Student Affairs dashboard.

- Dashboard numbers: upcoming events, events in the next 7 days, RSVPs for upcoming events and events at capacity.
- New event button.
- Fill bar for events with a capacity. It turns red at 100 percent.
- Full and Cancelled states are easy to spot.
- Switch between upcoming and past events.

Publish an event


Eventard

 Events
  **Dashboard**


Dr. Uwem Ekenyong
 Student Affairs
 

[← Dashboard](#)

Publish an event

It appears on every student's board as soon as you save.

1

Please fix the highlighted fields.

Title

eg. Career Fair 2026

2

Title must be at least 4 characters

Category

Choose a category

Pick a category

Venue

eg. Main Auditorium, Main Campus

Enter a venue

3

Starts

10/08/2026, 02:00 PM

Ends

10/08/2026, 04:00 PM

Capacity

Leave empty for no limit

Description

What will happen, who it is for and what to bring

Describe the event in at least 20 characters

Banner

Upload a photo (JPG, PNG or WebP, up to 3 MB) or pick one from the library.

 Upload image

4

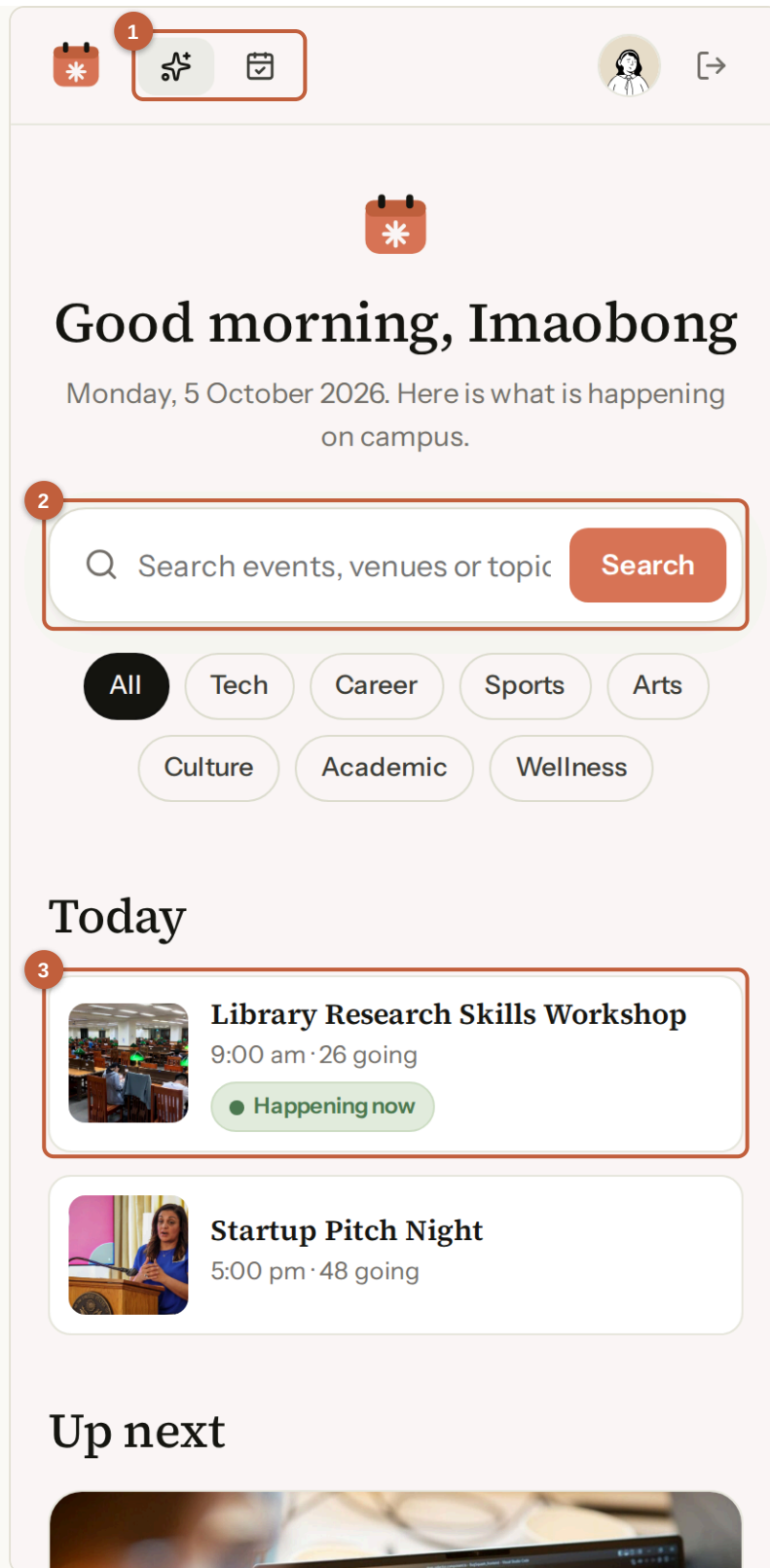


New event form after submitting it empty.

- 1 Summary message when the form has problems.
- 2 Inline field errors from the Zod schema, shown under each input.
- 3 Start and end use datetime-local inputs in campus time. Business rules reject past starts, ends before starts and events longer than 3 days.
- 4 Banner library of real photos. Admins can also upload a JPG, PNG or WebP up to 3 MB, which is stored in Postgres.

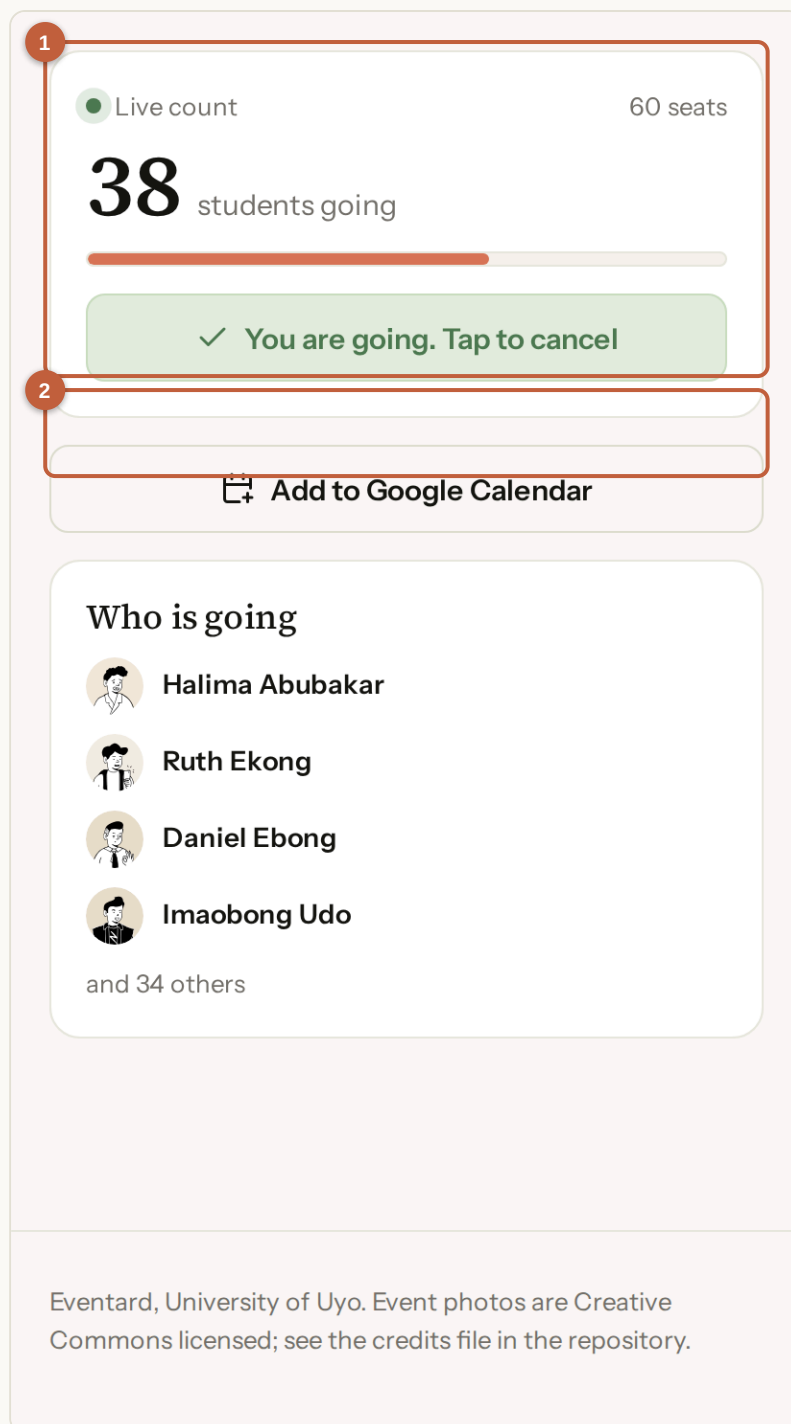
5. Mobile view

Layouts use CSS grid with `minmax(0, 1fr)` columns so long titles and tables never push the page wider than the screen. The mobile e2e test runs on a Pixel 7 profile and fails if the page scrolls sideways.



Board on a phone.

- 1 On phones the navigation collapses to icons.
- 2 Search stays full width.
- 3 Cards stack into one column; no sideways scrolling (checked in the mobile e2e test).



RSVP panel and calendar button on a phone.

- 1 On mobile the RSVP panel follows the event details.
- 2 Calendar button, full width for easy tapping.

6. Running locally

```
service postgresql start
sudo -u postgres createdb eventard
sudo -u postgres createdb eventard_test
cp .env.example .env          # set DATABASE_URL and SESSION_SECRET
npm install
npm run db:migrate
npm run db:seed                # resets and seeds demo data relative to today
npm run dev                    # http://localhost:3000
```

Script	What it does
npm run dev / build / start	Next.js development server, production build, production server
npm run start:prod	Migrate, seed only if the database is empty, then next start on 0.0.0 (used by Railway)
npm run db:generate	Create a new SQL migration after editing src/db/schema.ts
npm run db:migrate / db:seed	Apply migrations / reset and seed
npm test, test:unit, test:integration, test:e2e	Run the test suites
npm run docs:pdf	Rebuild this PDF

Set `EVENTARD_TODAY=2026-10-05` to freeze the clock for a rehearsal.

7. Testing

42

unit tests passed

12

integration tests passed

7

e2e tests passed

Suite	Tool	What it covers
Unit	Vitest	Event phases, RSVP rules, capacity helpers, campus day labels across UTC midnight, time validation, Google Calendar URL format and encoding, datetime-local conversion, Zod schemas, image checks, JWT signing and tampering.
Integration	Vitest + real Postgres (eventard_test)	Migrations and seed, RSVP toggle, full/past/cancelled refusals, unique constraint, 10 concurrent RSVPs for 3 seats, per-user scoping, search and category filters, admin stats and attendees, the end-after-start check constraint.
End to end	Playwright (desktop Chrome and Pixel 7)	Login redirect and prefill, wrong password, RSVP and live count, calendar link parameters, full event, inline form errors, admin publishes and a student's open board picks it up, mobile RSVP with no sideways scroll.

Latest run: unit 42, integration 12, 0 failed; e2e 7 of 7 passed (6 desktop, 1 mobile).

```
npm run test:unit
npm run test:integration # needs eventard_test
npm run test:e2e         # builds the app, seeds eventard_test with EVENTARD_TODAY=2026-10-05
```

8. Deployment

Eventard runs on Railway in the **school-projects** project as its own service with its own PostgreSQL database (**eventard-postgres**). The GitHub repository is connected to the service, so every push to main deploys.

Setting	Value
Build	npm run build
Start	npm run start:prod (migrate, seed if empty, next start -H 0.0.0.0)
Health check	/api/health (returns 503 if the database cannot be reached)
DATABASE_URL	\${{eventard-postgres.DATABASE_URL}}
SESSION_SECRET	random 48 byte value
NODE_ENV	production

These settings live in `railway.json` in the repository.

9. Five minute presentation script

Time	What to do and say
0:00	Problem: event news is scattered across WhatsApp groups and notice boards, so students miss talks and fairs. Eventard is one board with RSVPs and calendar reminders.
0:30	Open the sign-in page. Point out the prefilled demo login. Sign in as the student. Show the greeting, the Today strip and the Up next card.
1:15	Search for "hackathon", then click the Tech chip. Clear the filter. Point out Full and spots-left flags on cards.
1:45	Open "Code and Coffee". Say the count is 37. Click RSVP: it jumps to 38 and the button turns green. Explain the row lock that stops overbooking.
2:30	Click Add to Google Calendar. Google opens with title, time, venue and description filled in. Explain the link is built from the event data, dates in UTC, no API key.
3:00	Open My RSVPs to show only this student's bookings.
3:20	In a second window sign in as admin. Show the dashboard numbers and the red Full bar for Robotics Club Open Day.
3:45	Click New event, submit empty to show inline errors, then fill it in, pick a banner and publish. Switch to the student window: the new event appears within a few seconds without reloading.
4:30	Architecture in one sentence each: Next.js Server Components and Server Actions, Drizzle with PostgreSQL, JWT cookie sessions, tests (quote the counts), deployed on Railway with a health check.
4:50	Close: what we would add next (email reminders, QR check-in at the door). Take questions.

Image credits

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